

Project Omega: Distributed AI Orchestration

1. Executive Overview

Project Omega is a Tier-1 autonomous orchestration layer designed for heterogeneous compute clusters. It utilizes a Non-Euclidean Memory Mesh to manage state across 10,000+ edge nodes.

2. System Architecture

The core consists of the 'Aether Kernel' which handles sub-millisecond task dispatching. It is cross-linked with the 'Chronos Ledger' for immutable event sequencing. Security is enforced via the 'Hydra Protocol', a multi-sig biometric validation layer.

3. Inter-agent Communication

Agents communicate using the 'Singularity Bus'. When Agent-01 (Primary) initiates a 'Neural Handshake', Agent-07 (Watcher) must validate the checksum against the 'Deep-State Vault'. If mismatch occurs, 'Protocol-99' is triggered, resulting in immediate cluster isolation.

4. Technical Requirements

Minimum latency: 0.4ms. Throughput: 1.2M OPS. Redundancy: 3x mirror via 'Shadow-Node' synchronization.